



Lecture 24:

Introduction to Signal Integrity and Power Delivery

Contents



This Topic: Introduction to Signal Integrity and Power Delivery

- Introduction
- Transmission Line Effects
- Cross Talk
- Ground Bounce and Power Supply Noises
- Current Return Path
- Power Delivery Network

Introduction

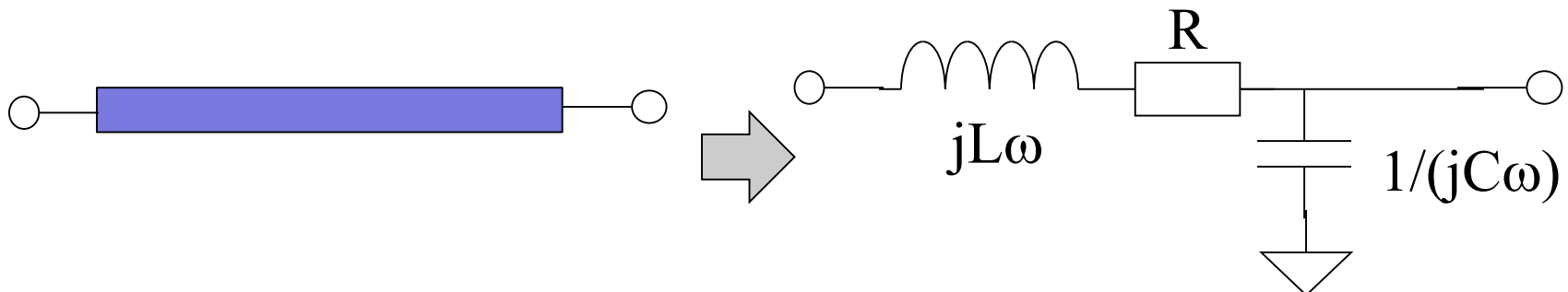


- Signal Integrity (SI) deals with the issues in signal quality
 - i.e. SNR in time-domain systems
- SI issues are primarily seen at PCB level and now an increasing concern within VLSI on-chip circuits
- SI issues introduce pattern-dependent and/or soft failures, which are difficult to debug
 - Wrong data captured (poor data signal)
 - Data captured at wrong time (poor clock signal)

Introduction



- SI problems are related to parasitic effects of signal path interconnects:
 - Resistance Effects (only effect at low frequency)
 - Capacitance Effects (should be added at mid-frequency)
 - Inductance Effects (must be added at high frequency)
 - Antenna Effects (must be added at very high frequency)



Introduction



- Power Delivery (PD) problems are related to quality in power supply
 - Also known as Power Integrity (PI)
- The parasitic effects of power supply interconnects:
 - dI/dt noise
 - IR drop
 - Good power (on/off) response
 - Startup time
 - Overshoot
 - Undershoot

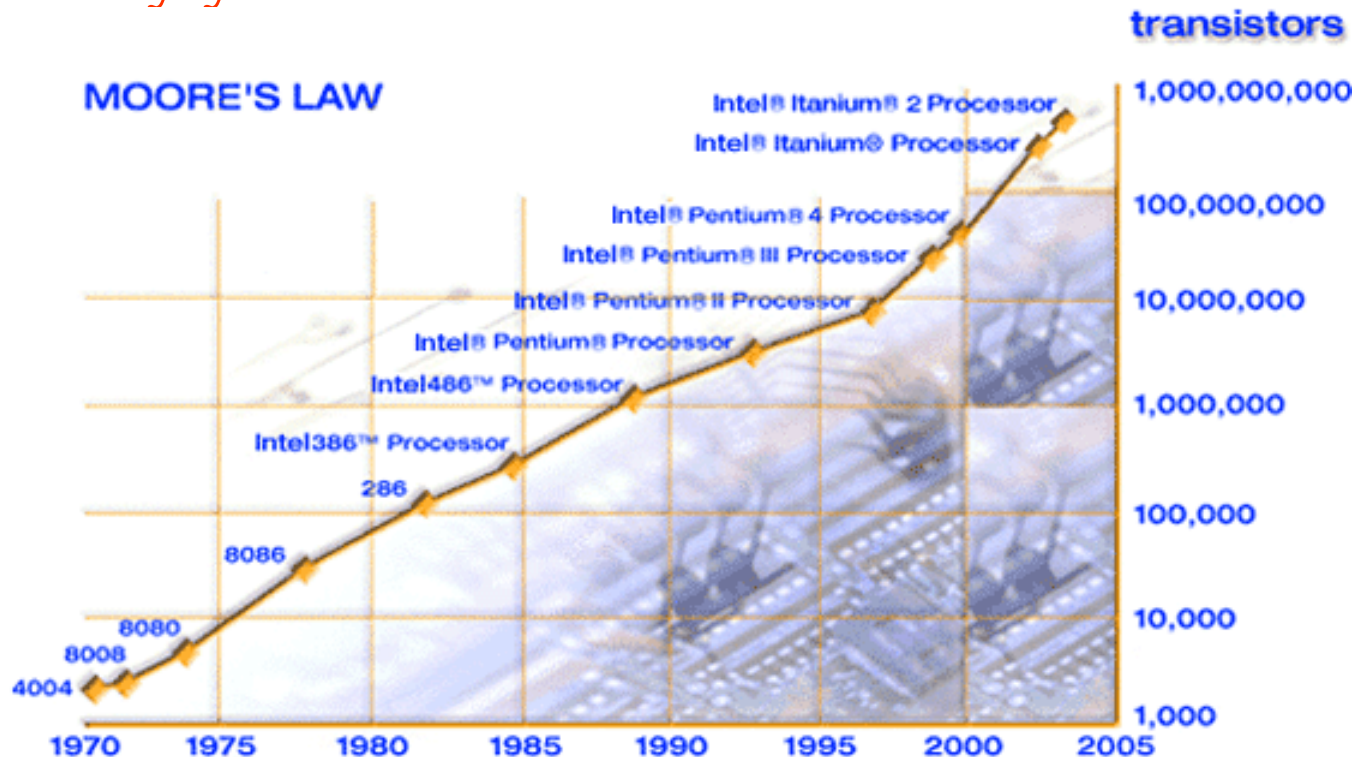
Signal Integrity Design Approaches



- Frequency-Domain Approaches:
 - Impedance matching of T-Line at the package interfaces
- Time-Domain Approaches:
 - Package selection
 - Limitation on simultaneous switching output
 - Limitation on switch speed
 - Supply Decoupling

Why Everything Is Getting Harder: Moore's Law

- Gordon Moore, chairman emeritus of Intel, predicts the “doubling of transistor density on a manufactured die every year”... *in 1965*



Source: <http://www.intel.com/technology/silicon/mooreslaw/index.htm>

National Technology Roadmap for Semiconductors



Year	Smallest Feature, μm	Dynamic RAM		Microprocessors			Wiring levels per chip	ASIC Package pins/balls
		Chip size, mm^2	Billions of bits per chip	Chip size, mm^2	Millions of transistors per cm^2	Chip-to-board clock, MHz		
1997	0.20	280	256M	300	10	250	6	1100
1999	0.14	400	1G	340	17	480	6-7	1500
2001	0.12	445	*	385	29	785	7	1800
2003	0.10	560	4G	430	50	885	7	2200
2006	0.07	790	16G	520	110	1035	7-8	3000
2009	0.05	1120	64G	620	240	1540	8-9	4100

Source: Semiconductor Industry Association; 1997

Key Signal Integrity Issues

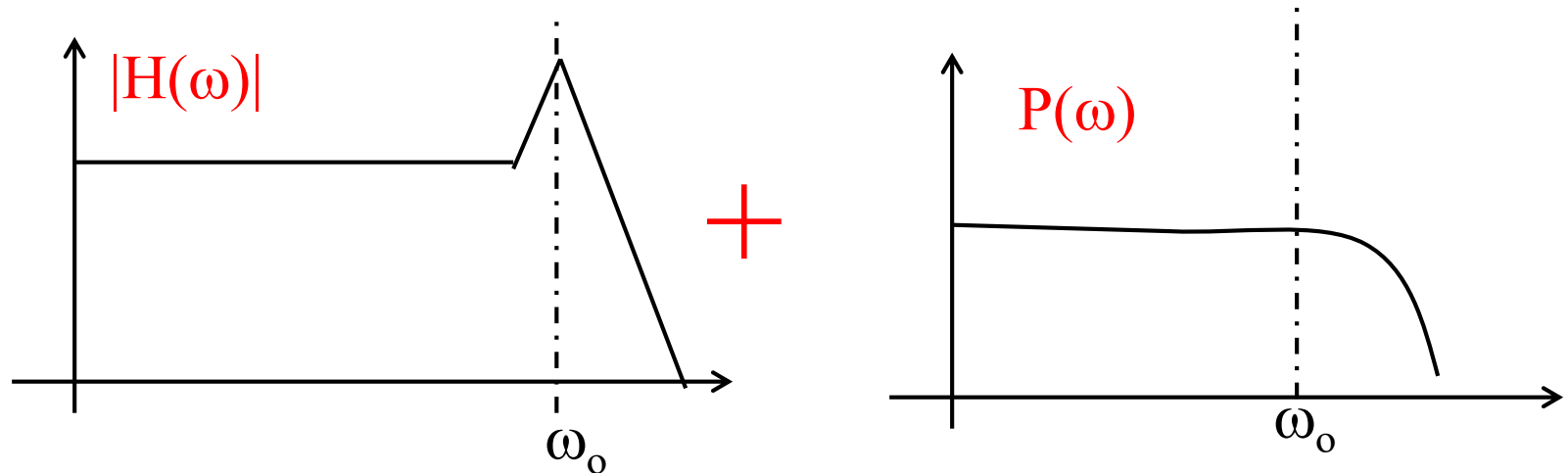


- Four Major Types of SI Issues
 - Ringing (reflection) Effects
 - Delay (jitter) Effects
 - Crosstalk (noise & jitter) Effects
 - dI/dt (and IR) Noise (& jitter) Effects

Ringling & Reflection Effects



- Ringing (reflection) occurs in LC based circuits, channel, or line that meet **Both conditions**:
 1. Signal TF has peaking response at certain frequency
 2. Signal has energy at the peaking frequency



Ringling & Reflection Effects



- Solutions to Ringling & Reflection Effects

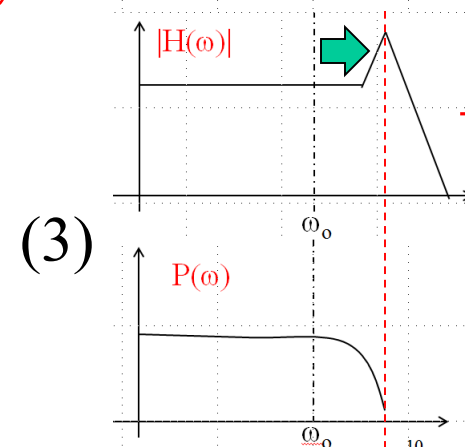
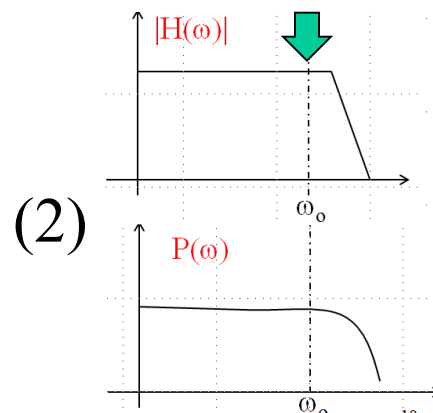
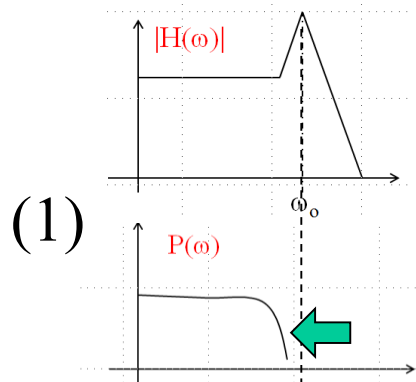
1. Slew Rate Control

- Get rid of energy at ring frequency

2. Proper Termination

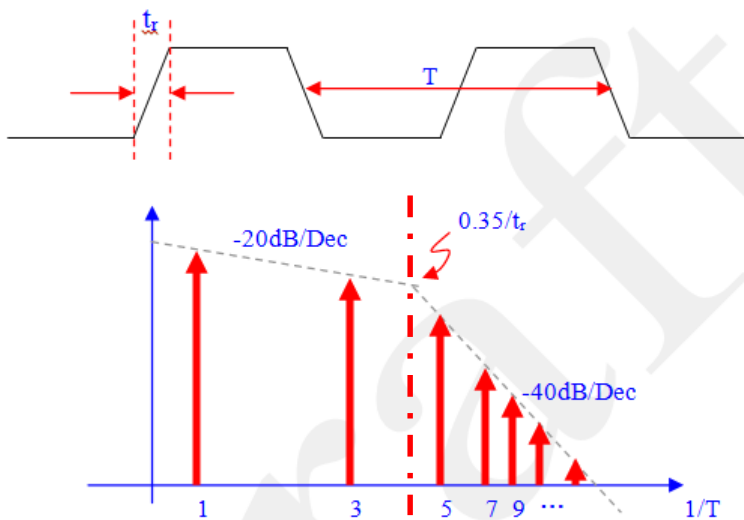
- Get rid of peaking response

3. Push ω_o out of signal energy band (if you can)



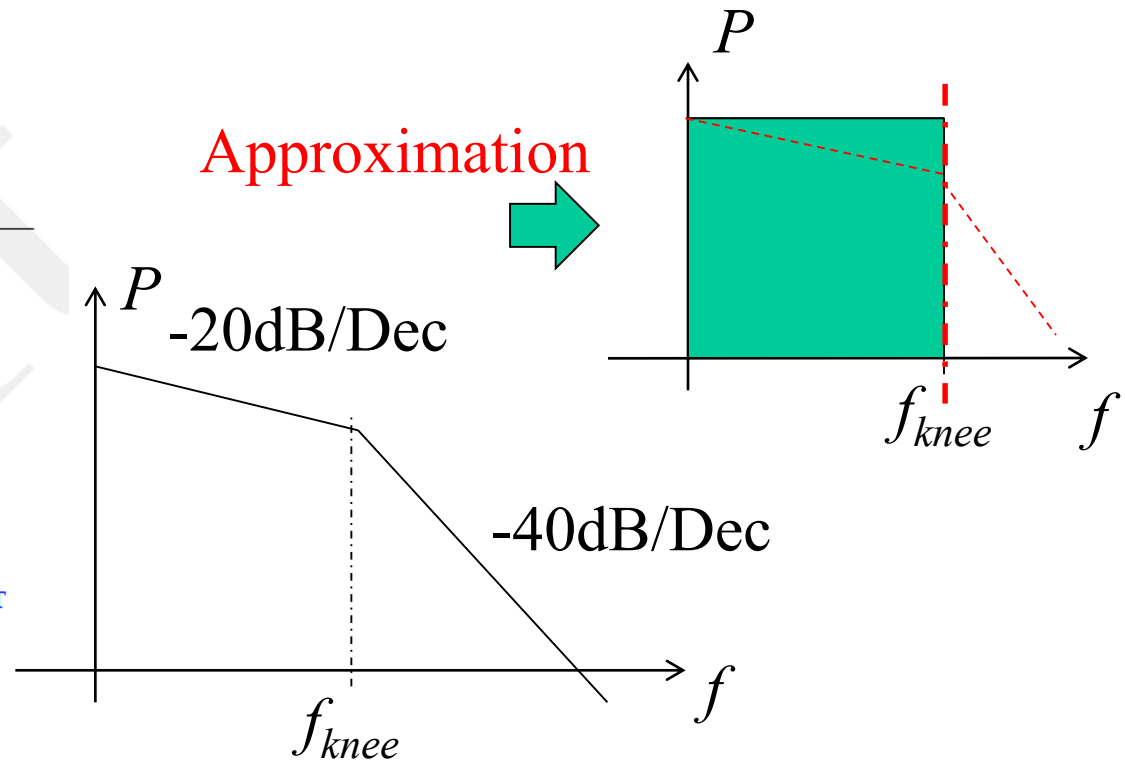
The Knee Frequency

- For digital signal, power density in frequency band higher than the Knee Frequency are usually highly attenuated.



$$f_{\text{knee}} \sim 0.35/t_{\text{rise}}$$

Approximation



Design Rule: Ringing & Reflection Effect

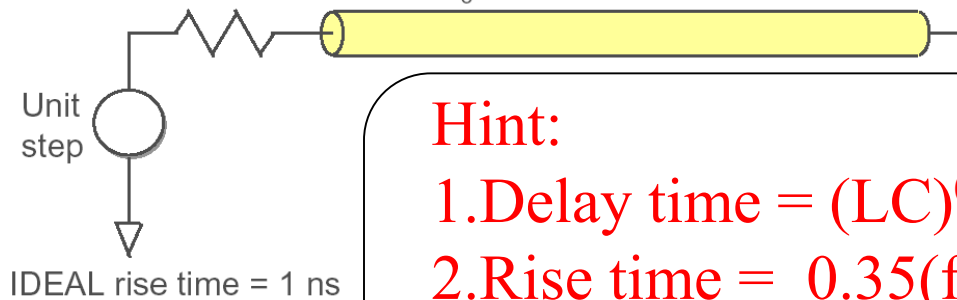


- Classic termination rule of thumb

– When T-line delay exceeds 1/6 of rise time, termination is required to control ringing

Source impedance:
Line Driver = 10 ohms

Lossless, distortionless transmission line
 $Z_0 = 70$ ohms



Hint:

1. Delay time = $(LC)^{0.5} = (\omega_o)^{-0.5}$

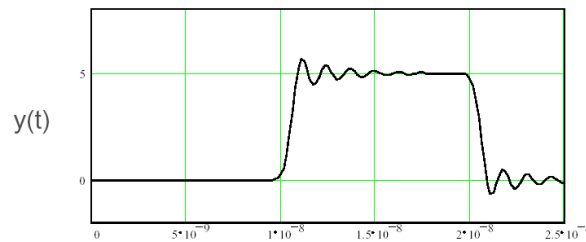
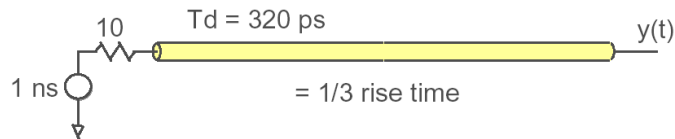
2. Rise time = $0.35(f_{knee})^{-0.5} \sim (\omega_{knee})^{-0.5}$

3. $\rightarrow$ significant energy @ peaking $\rightarrow$ need termination

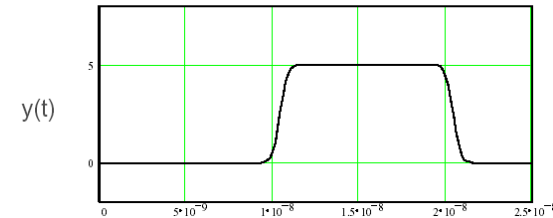
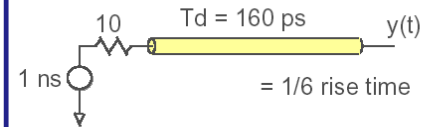
Example: Ringing & Reflection Effects



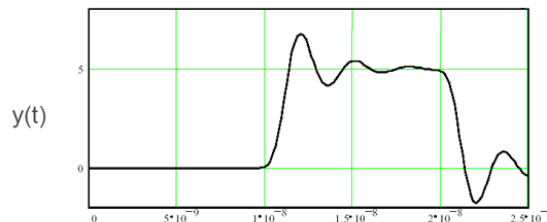
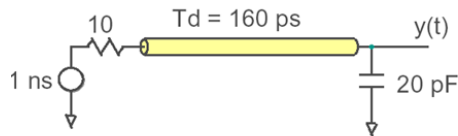
1. Long Line & Poor Termination



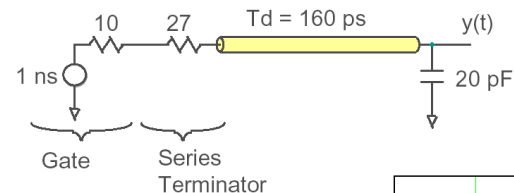
2. Short Line & Poor Termination



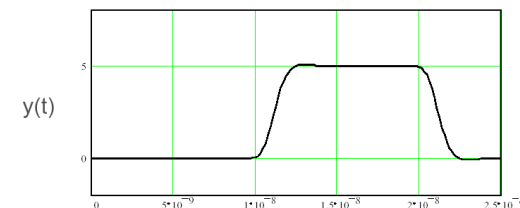
3. Effectively Long Line & Poor Termination



4. Long Line & Good Termination



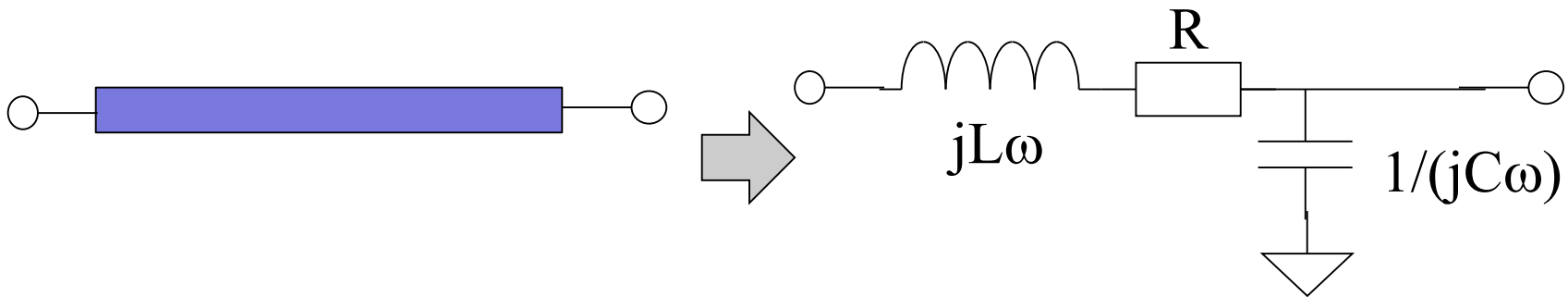
Even at short lengths, terminators are sometimes necessary.



Delay Effect



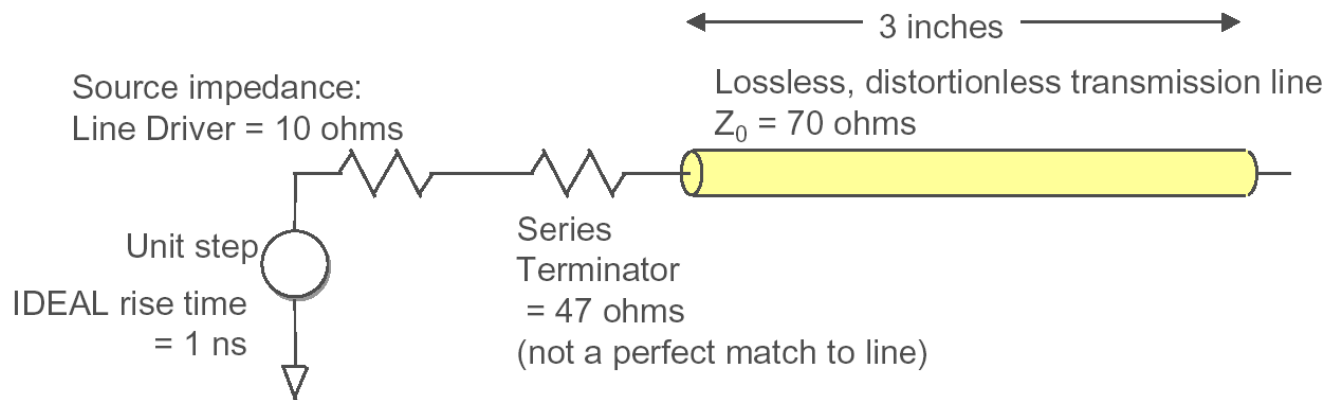
- Delay effect of a LRC trace is caused by time constant to charge/discharge the line
 - $t \sim (LC)^{0.5}$ (high Q) or
 - $t \sim RC$ (low Q)



Design Rule: PCB Delay Effects



- Classic rule of thumb for delay
 - The delay of an inside (stripline) trace in FR4 PCB is typically 180 ps/inch



Delay Effects

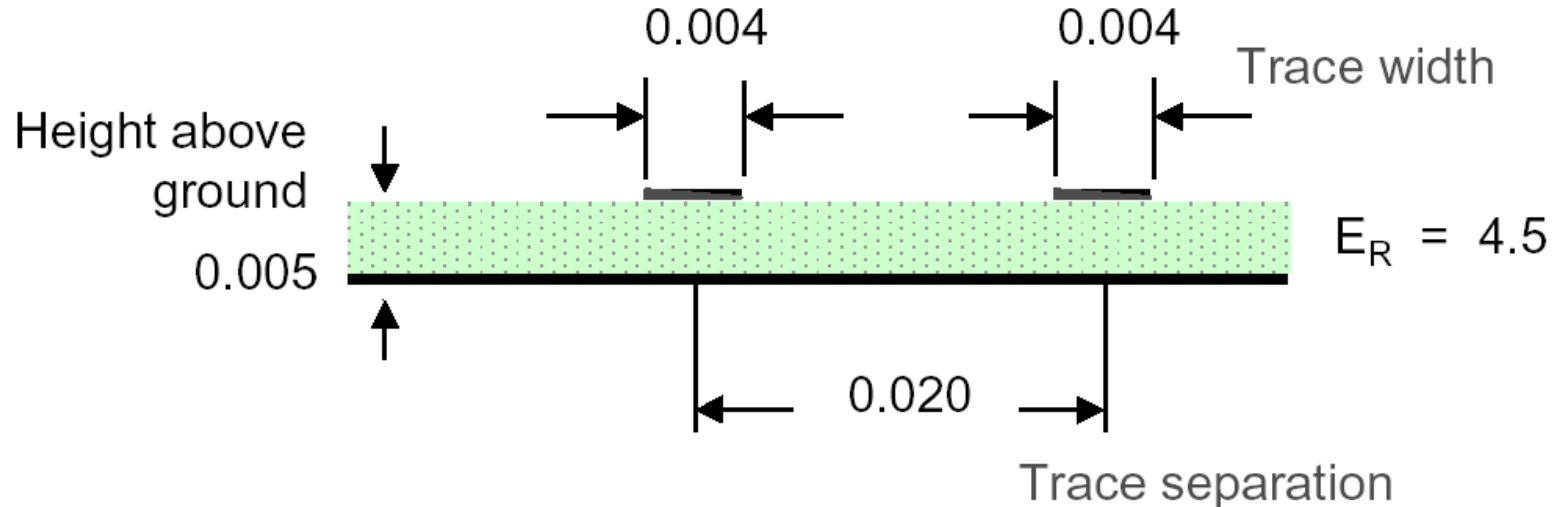


- The actual line delay depends on:
 - Line length
 - Rise time
 - Connected loads
 - Observation point (end or middle)
 - Logic thresholds

Design Rule: Crosstalk Effects



- Common crosstalk rule of thumb
 - Maintain spacing between traces of 4:1 (horizontal spacing to vertical height above ground)



Crosstalk Effects



- PCB designers care about crosstalk because it can cause the following performance problems:
 - Increased noise levels
 - Unplanned spikes
 - Jitter on data edges
 - Reflections of undesired signals

Crosstalk Effect Modeling

- Crosstalk begins as a digital gate outputs a rising edge to the aggressor line, which starts to propagate along the line:
 - Both the mutual capacitance, C_m , and the mutual inductance, L_m , couple or "crosstalk" a voltage into the adjacent victim line.
 - The crosstalk voltage appears on the victim line as a narrow pulse with a width equal to the pulse rise time on the aggressor line.
 - On the victim line, the pulse divides into two pulses, and each starts to propagate in opposite directions, effectively dividing crosstalk into two parts:

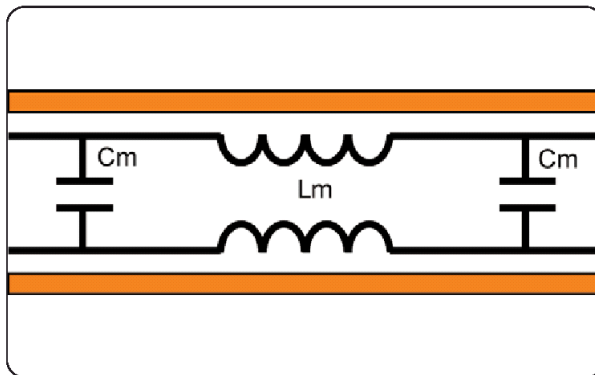


Figure 1. Mutual Impedance between lines on PCB

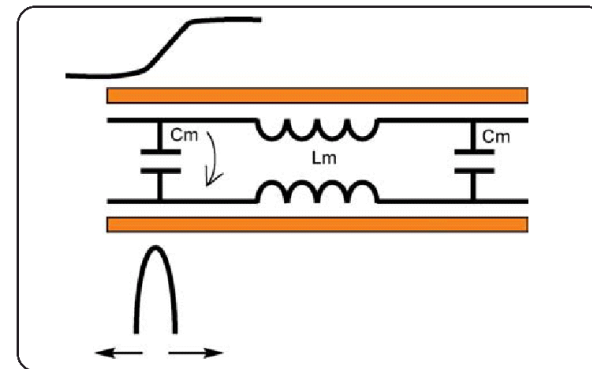
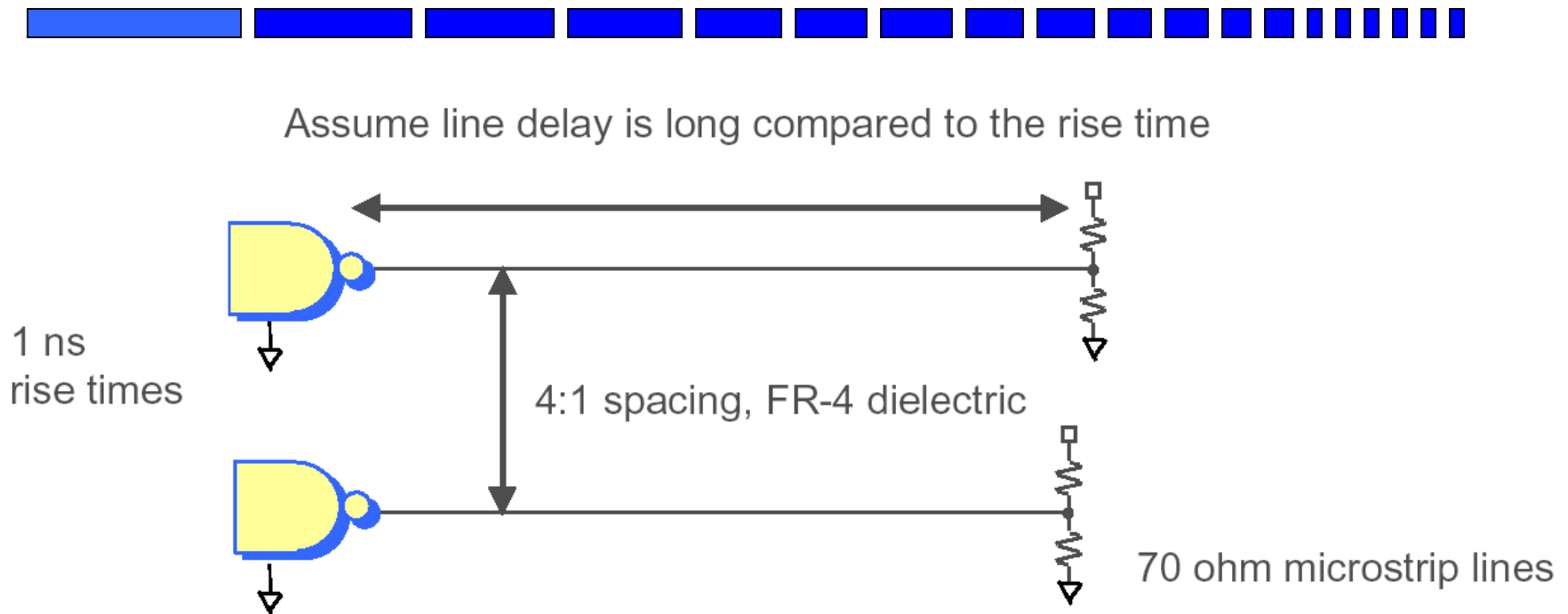


Figure 2. Capacitively coupled crosstalk

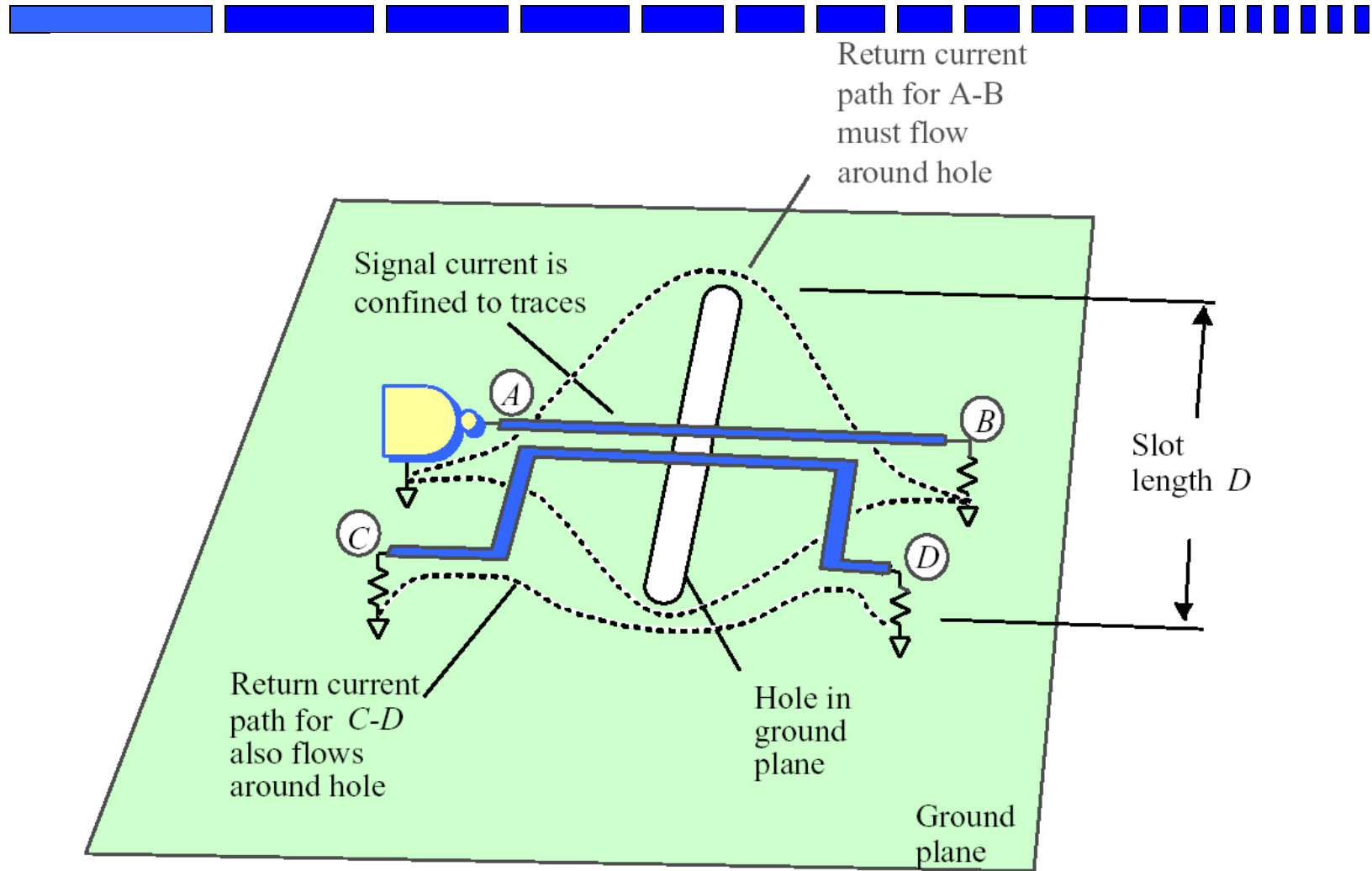
Crosstalk Effects



Expect about 5-6 % crosstalk on the longest lines.

Crosstalk should vary inversely with the square of trace separation.

Crosstalk Over Hole in Reference Plane

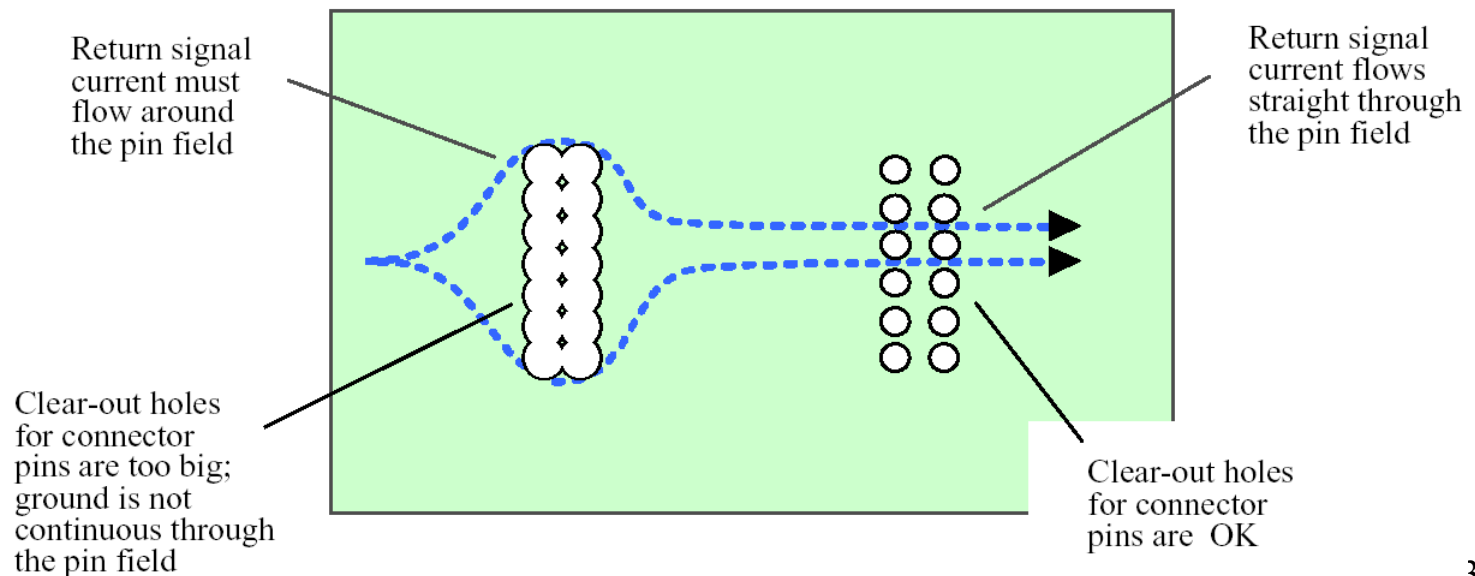


Crosstalk soars to ridiculous proportions.

Crosstalk at Connector Positions



- Reference-plane slots happen on dense back planes which pass through fields of connector pins.
- Always make sure the solid-plane clear-outs around each pin maintain continuity of the plane between all pins.



Cross-Talk Modeling Approaches



- Frequency-domain techniques:
 - Measure the clock harmonics located in the frequency spectrum relative to the Spec (EMI and others) limit levels at these frequencies.
- Time-domain techniques:
 - Rise/fall times (e.g. 10% to 90%);
 - Knee Frequency calculation based on Rise/fall times
 - Xtalk level based on frequency response.

Crosstalk Effects



- Actual crosstalk depends on:
 - Line length
 - Rise time
 - Line spacing
 - Connected loads
 - Trace orientation (same way or opposite)
 - Reference plane geometry

dI/dt Noise and Ground Bounce

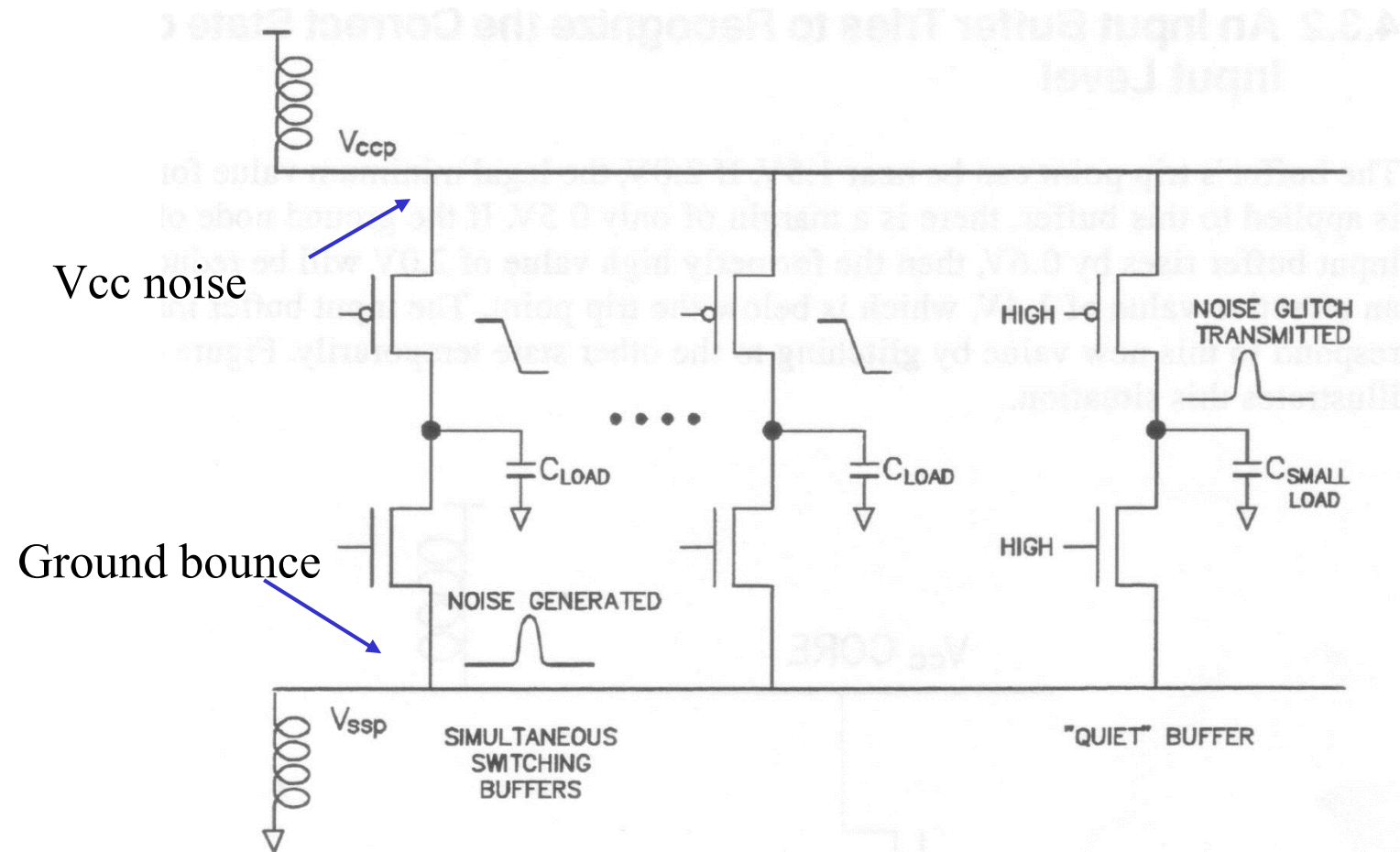


Figure 4–1: An Asynchronous Quiet Output Buffer Sharing A Noisy V_{SSP} Bus

dI/dt Noise and Ground Bounce

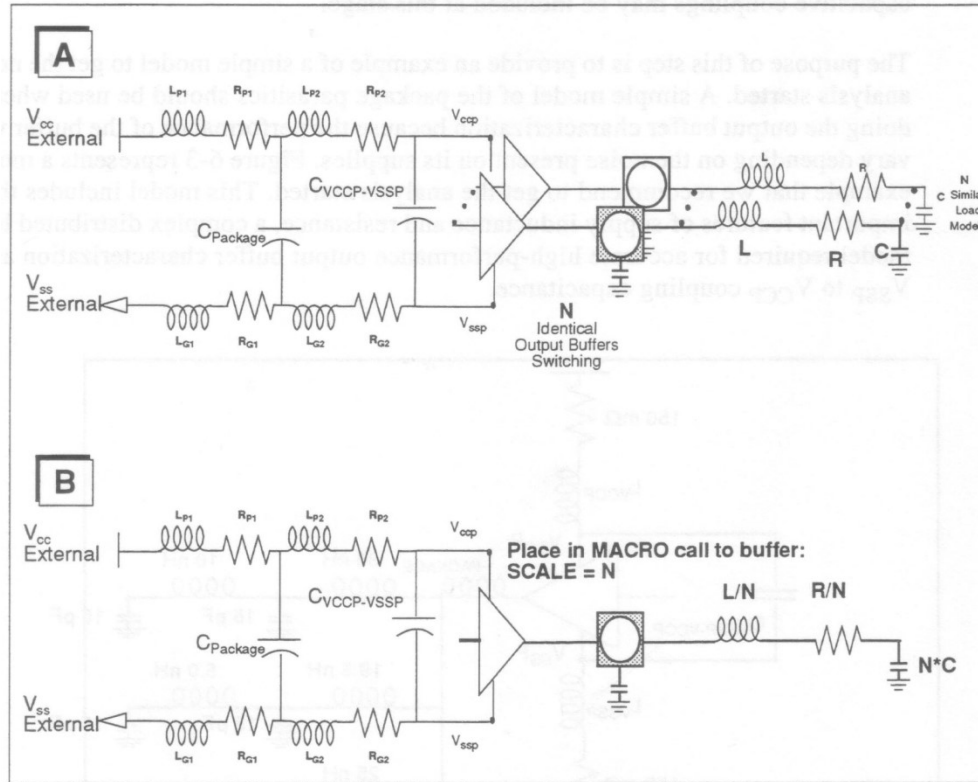


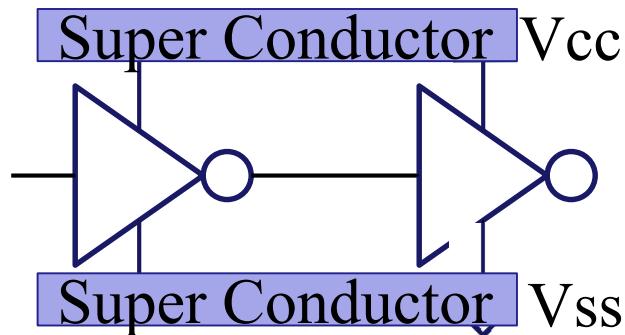
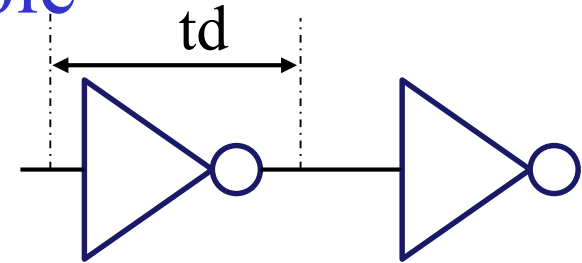
Figure 6-4: Simplified Simulation of N Buffers Switching Simultaneously

Noise simulation results of N identical-output buffers switching together (Figure 6-4) is equivalent to the results of one output buffer switching where the buffer has been scaled by N , and N loads are attached in parallel.

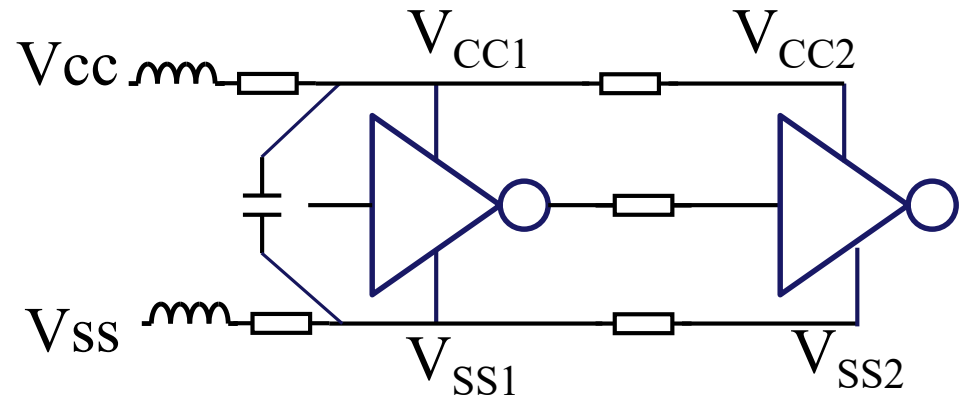
Why PD/PI Design?

- Gate Delay modeling Example

- Super Conductor Model
- Realistic Model



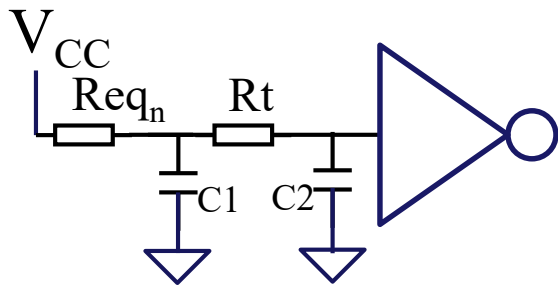
(a) Ideal (Super Conductor) Model



(b) More realistic Model

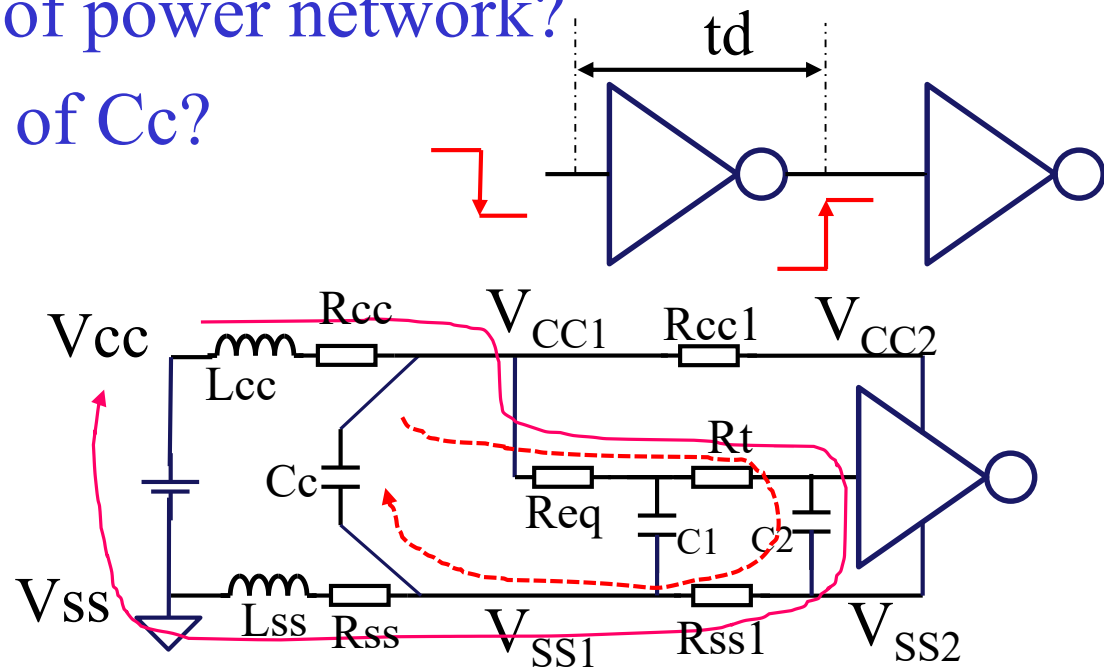
What is the Impact of PD Network?

- Current paths in 0→1 transition
 - What's the effect of power network?
 - What's the Effect of C_c ?



$$t_d = 0.7(R_{eqn}(C_1 + C_2) + R_{int}C_2)$$

(a) Ideal (Super Conductor) Model

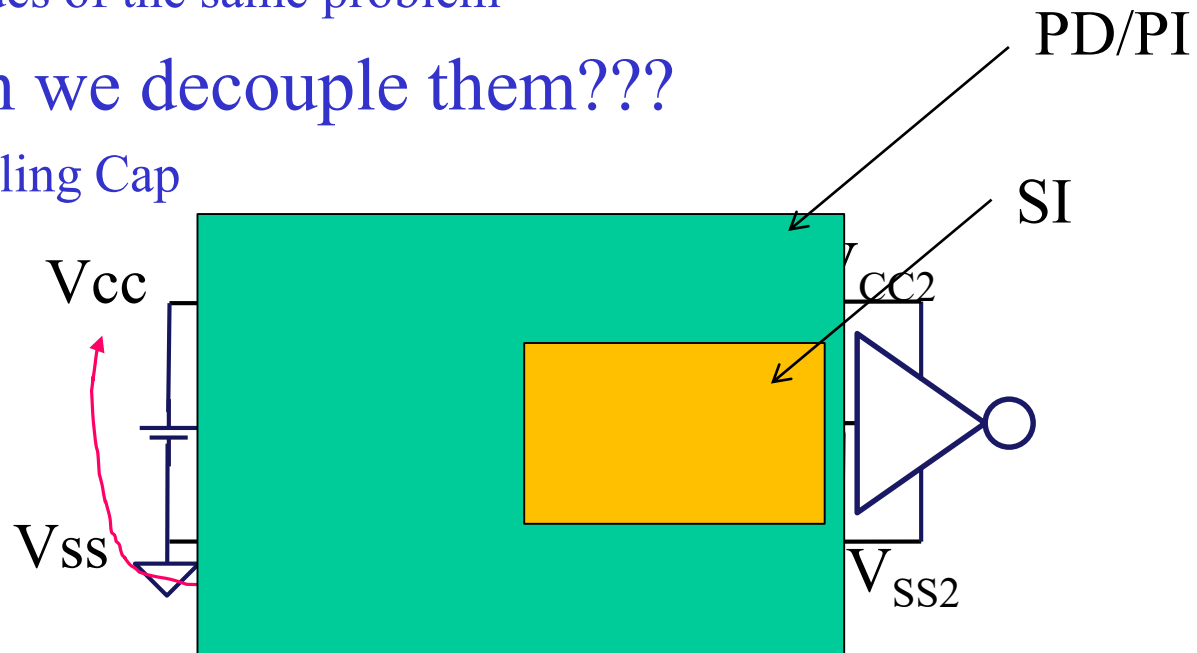


$$t_d = ???$$

(b) More realistic Model

SI and PD Design Partition

- Power Network → PD Design
- Signal Path → SI Design
- However, they are interrelated!!!!
 - Two sides of the same problem
- How can we decouple them???



Power Delivery Simulation

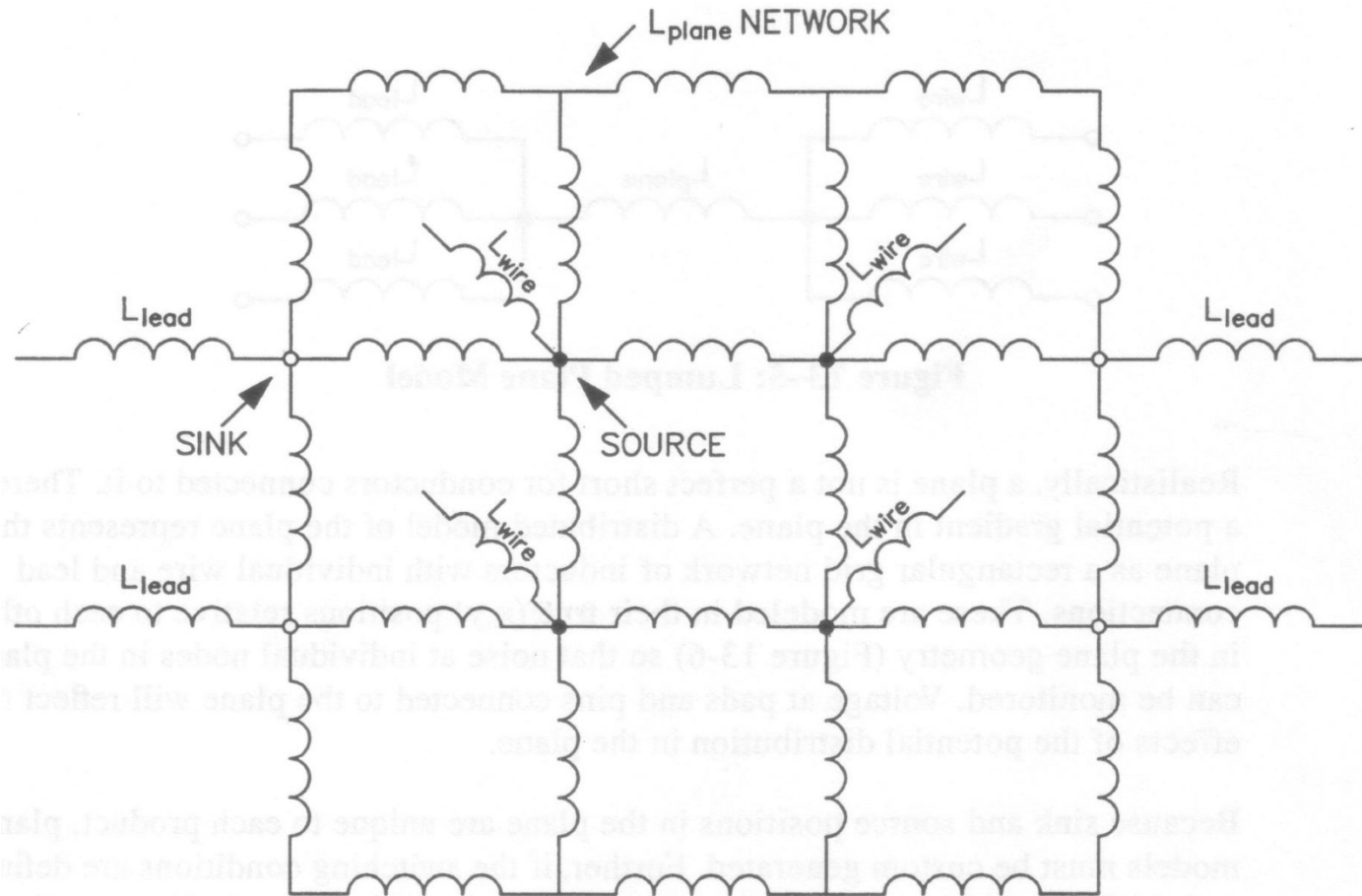


Figure 13-6: Distributed Plane Model

Supply Decoupling Capacitor

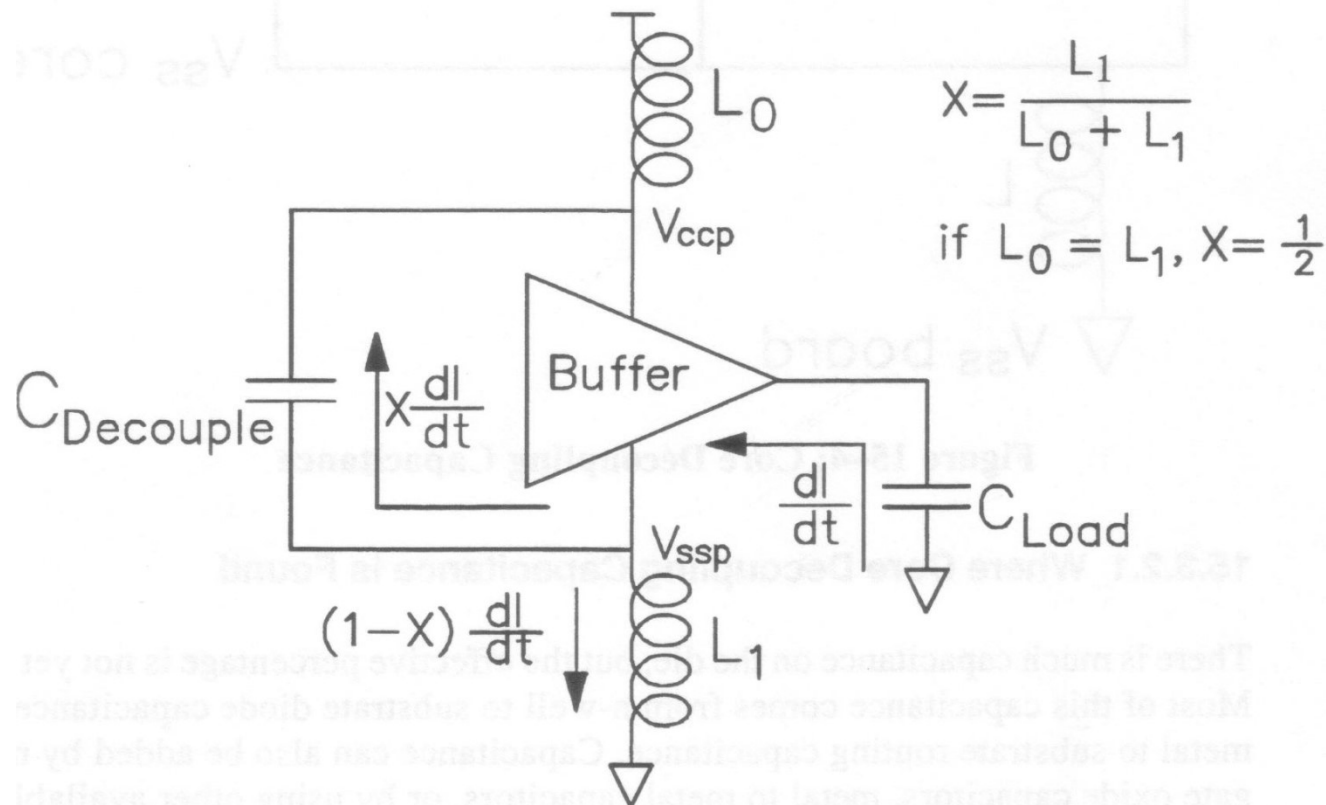
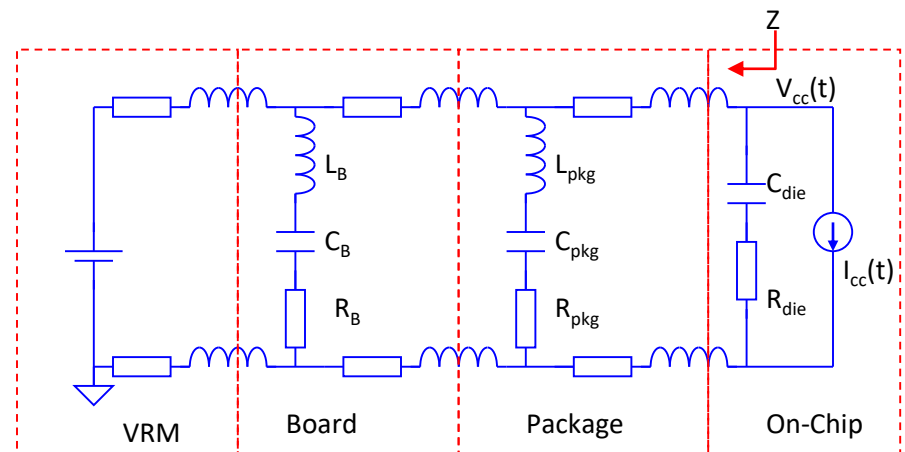
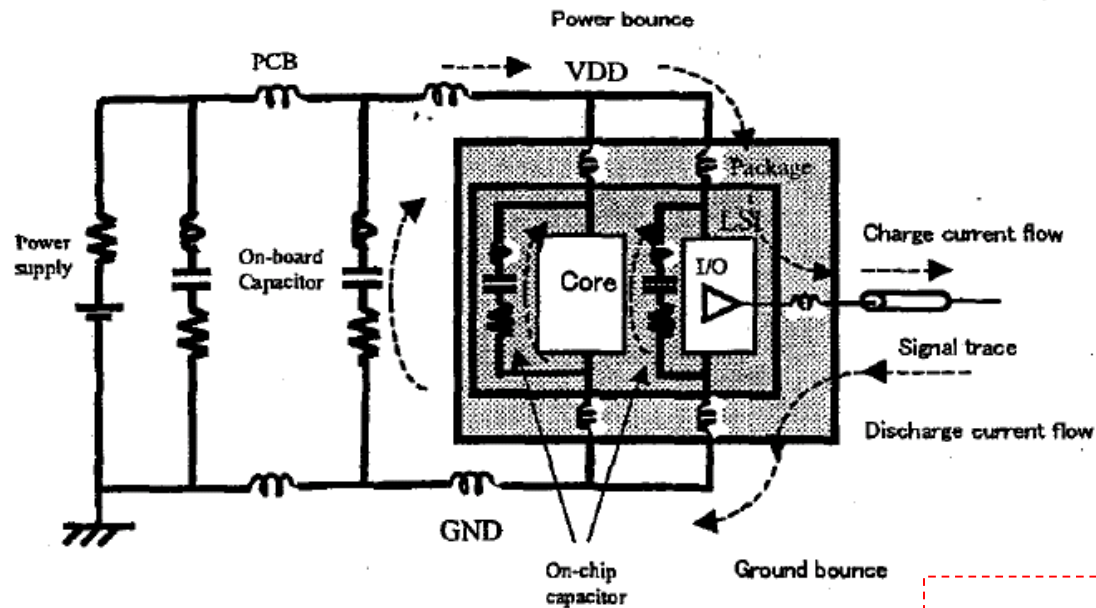


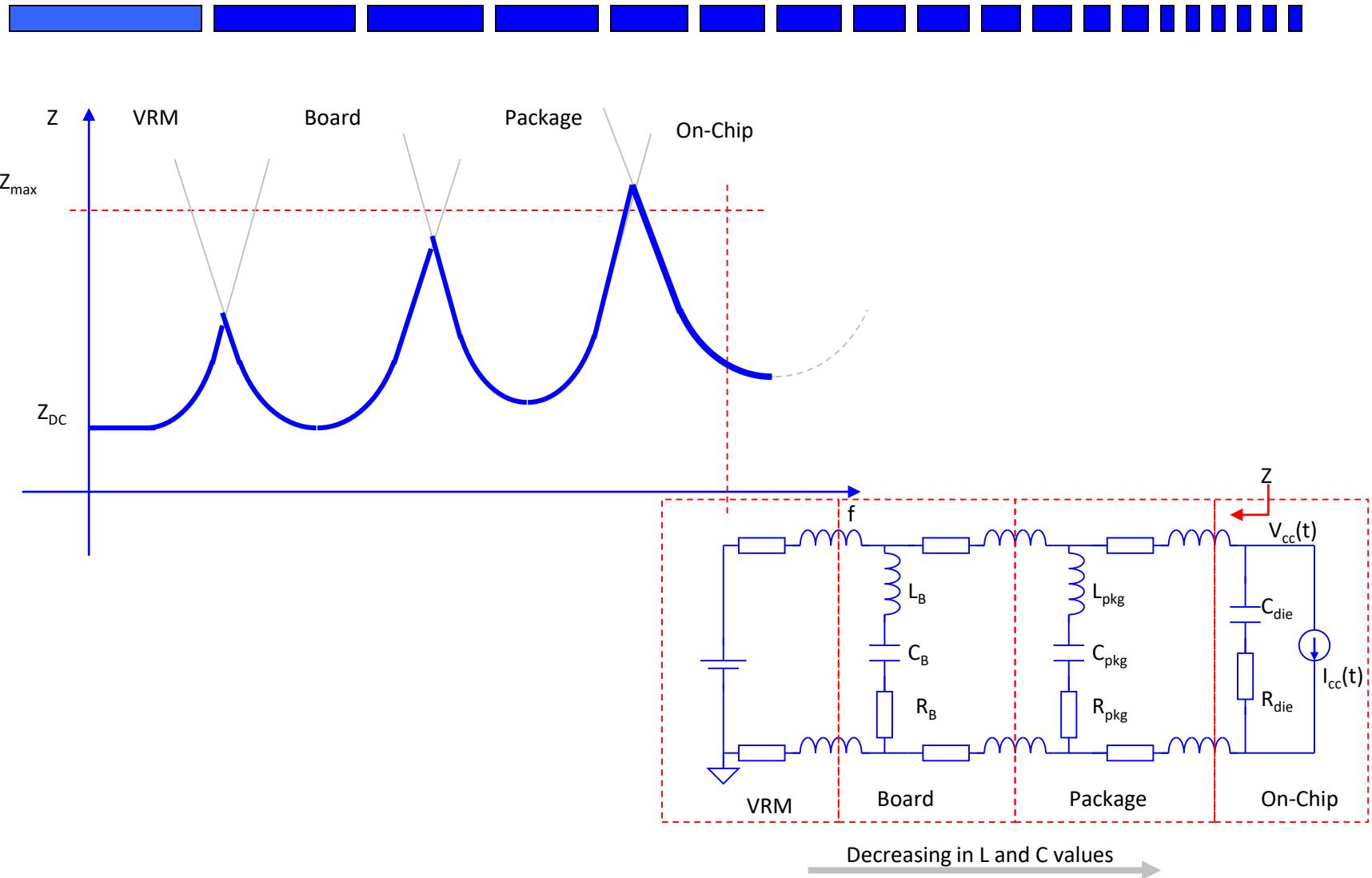
Figure 15-3: Output Buffer Power Supply Decoupling Capacitance

Equivalent Power Delivery Network



Decreasing in L and C values →

Power Delivery Network Impedance Profile



Reading Assignment



- Please read the related Chapter of the textbook